EDES 301

MyoFX Proposal

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MyoFX: EMG-Based Expression Control for Guitar Effects

I want to build a wearable biosignal controller that uses muscle activity to control guitar effects in real time.

The first prototype uses a MyoWare EMG sensor to measure forearm muscle activation. The EMG signal is read by a PocketBeagle, converted into a normalized expression value, and sent to a Daisy Pod, which performs the real-time guitar audio processing.

MyoWare was selected because it provides an integrated EMG front end and simple analog outputs, which reduces the hardware complexity of the first prototype.

(If MyoWare is considered too expensive, we can find other good options to replace it)

Background Information:
MyoWare Hardware applications

https://github.com/AdvancerTechnologies/MyoWare-Arduino-Library (official library)

https://github.com/melissayu03/myoware-emg-detector (hardware signal acquisition)

https://github.com/Araadh3111/Bionix (EMG to servo control)

Background Information: EMG to Music

https://www.hackster.io/upsidedownlabs/muscle-melody-play-music-with-your-muscle-movements-emg-c0a0c0

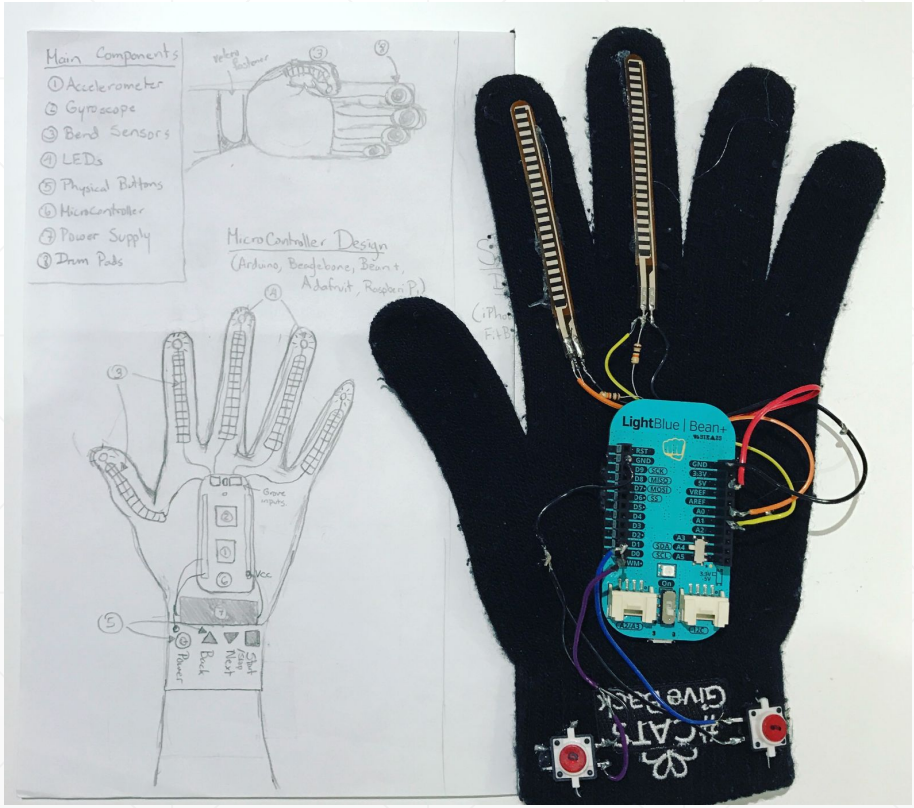
https://nime.org/proceedings/2025/nime2025_35.pdf



Background Information: Flex Bend

https://github.com/interactiveaudiolab/DataGlove

https://github.com/sneak-thief/Sneaky-Gestures-V2





Background Information: DaisySeed community

https://community.daisy.audio/t/esp32-control-of-daisy-seed/1226 (how others build communications with other MCUs as DaisySeed's input)

https://community.daisy.audio/t/daisy-serial-interface/1300/3 (serial problems)

https://github.com/bkshepherd/DaisySeedProjects/tree/main/Software/GuitarPedal (examples of MIDI CC input controlling DaisySeed)

Improvement: From Multi-Parameter Controllers to One Playable Tone Macro

Existing projects already show that the main parts are possible:

- MyoWare can measure muscle activity and turn it into a control signal.
- EMG music projects like EMMA show that muscle signals can be used for musical interaction.
- Daisy Seed projects show that external controls such as MIDI CC or knobs can change DSP parameters in real time.

But many of these systems use several sensors to control several different parameters.

For MyoFX, I want to use a simpler control architecture. I use one EMG signal to represent one overall expression level. That value can then change several effect parameters together instead of making the player control each one separately.

Improvement: From Multi-Parameter Controllers to One Playable Tone Macro

The main improvement is not adding more controls, but reducing the control complexity. Instead of asking the player to control several effect parameters separately, I want to use one EMG signal as one overall expression value and let that value change the guitar tone in a coordinated way.

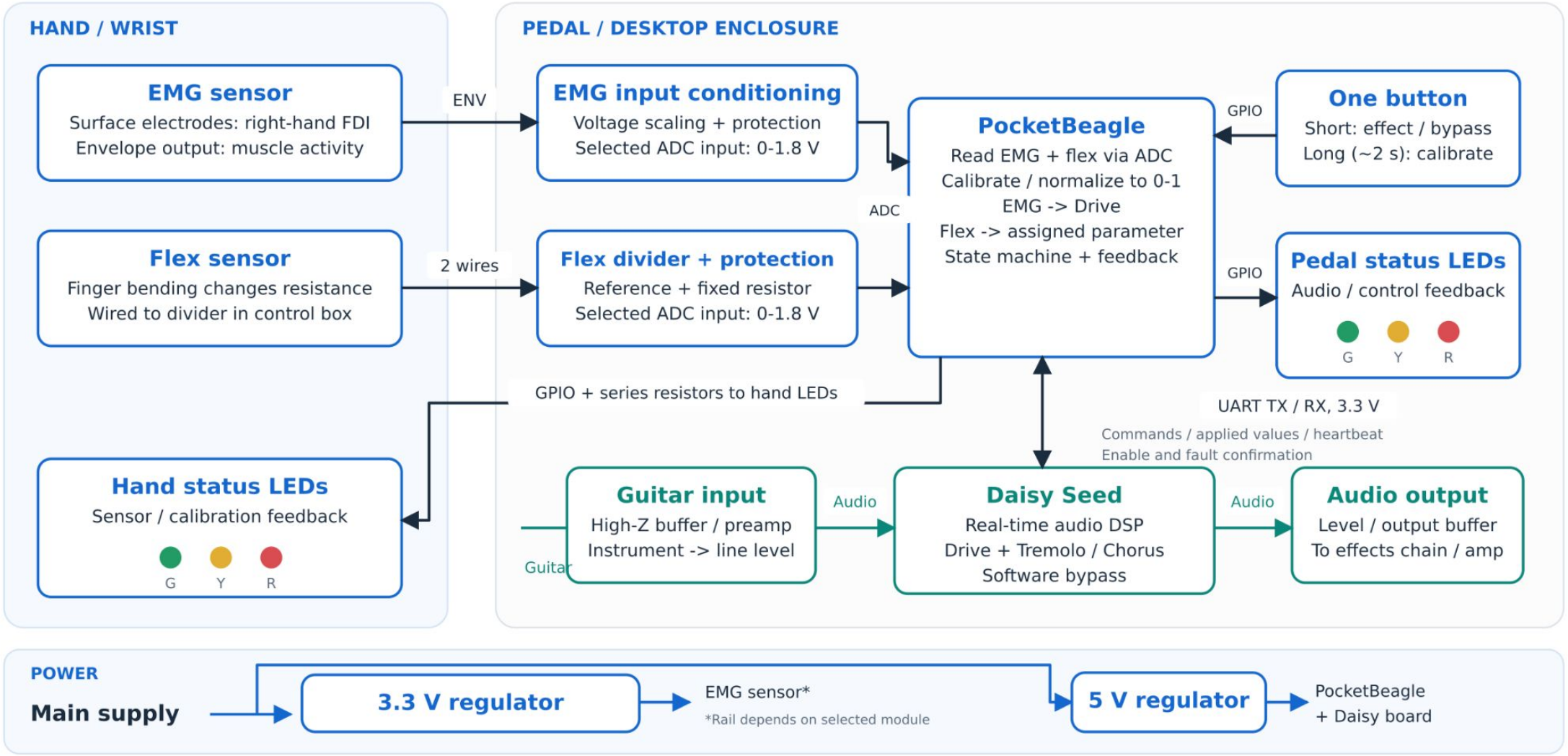
The first goal is to test whether physiological input can become a predictable and playable musical interface, similar to how an expression pedal gives continuous control.

If this works, the next version can improve the system in two ways:

- add Bluetooth so the sensing side can become wireless and more wearable

- add finger flex as a second control dimension for richer expression

Prototype to check feasibility



A state machine on the PocketBeagle manages startup, calibration, active effects, bypass, and fault states. At startup, the system remains in bypass until the user holds the button to calibrate the EMG and flex inputs. Once calibration is complete, a short press toggles between active effects and bypass while retaining the calibration data for the current power session. During operation, the PocketBeagle maps sensor readings to effect parameters and sends them to Daisy, using Daisy's feedback to update the status LEDs. Brief sensor disturbances cause the system to hold the last stable parameter values, preventing sudden changes in tone. Persistent disturbances or a communication heartbeat timeout trigger a fault indication. Once valid signals remain stable, control can resume using the existing calibration; the user can hold the button again whenever recalibration is needed.

$20

Components / Budget

Component?	EDES301 to Buy?	Cost
MyoWare 2.0 https://www.sparkfun.com/myoware-2-muscle-sensor.html	no	$42.95
Daisy Pod	no	$79
https://www.amazon.com/Smartmiao-SZ-3pcs-Muscle-Sensor/dp/B0G5JQYTN4/ref=sr_1_1?crid=2T4C8S0ASD6O2&dib=eyJ2IjoiMSJ9.7S2VZ3qEbV6Mn59QSw6knI7s7RF7-BG3XqvUG_7NIG7GjHj071QN20LucGBJIEps.uXCTmHX02yHq5fzJs42jsgcP-9P7PznBFOIB19okYWw&dib_tag=se&keywords=3pc+muscle+sensor+kit&qid=1791601450&sprefix=%2Caps%2C131&sr=8-1	yes	$22.58/3
Flex Sensor 2.2" https://www.sparkfun.com/flex-sensor-2-2.html	yes	$12.50
(if enough time after finished)BlueTooth device	no	?
(if enough time after finished)ESP32 Feather https://www.adafruit.com/product/5900 LiPo Battery	no	$20